

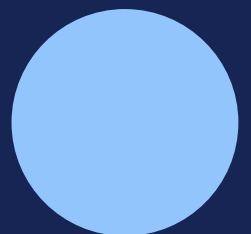
AIAA3102 FINAL PROJECT / TOPIC B

Dataset Forensics and Label Audit

An evidence-calibrated audit of duplication, cross-split leakage, likely label errors, shortcut features, and annotation ambiguity in the UCI SMS Spam Collection.

Fixed course split: 4,460 train / 1,114 held-out

Reproducible CPU pipeline | July 2026



Executive Summary

A simple word TF-IDF classifier reaches 99.10% accuracy and 96.55% spam F1 on the fixed held-out split. This project tests whether that score supports a trustworthy generalization claim. The answer is qualified: the model is strong, but the dataset contains repeated templates, cross-split leakage, likely label-policy errors, and highly predictive shallow cues.

The main validity result: removing 63 held-out rows linked to accepted exact or near training matches changes overall accuracy by only 0.05 points, but lowers spam recall from 93.96% to 90.91%. Accuracy alone hides the audit's largest effect.

The audit finds 290 exact-duplicate clusters involving 705 rows, including five clusters crossing the fixed split. At the public 0.92 TF-IDF cosine threshold, 365 non-exact candidate pairs remain; the 348-pair review queue accepts 200 pairs and rejects 148 false positives. Eighty-five accepted pairs cross the split and affect 60 unique held-out rows. Fifteen likely label errors are reported with independent model and policy evidence, while fourteen ambiguity claims distinguish genuine policy boundaries from model uncertainty.

All eleven shallow features reach 98.20% accuracy and 93.15% spam F1. They are useful task cues but fragile: legitimate price discussions become false positives, and spam without numeric markers can become false negatives. Training-side cleaning lowers the public benchmark, while adjudication-aware evaluation raises it by excluding four disputed held-out rows. The latter increase is partly circular, so intervention results must be interpreted by mechanism rather than by direction alone.

Key outputs

Output	Result
Ranked audit	36 claims; six per issue type; 52.8% high confidence
Duplicate audit	290 exact clusters; 5 cross-split exact clusters
Leakage sensitivity	63 held-out rows excluded; spam recall -3.05 percentage points
Label review	15 likely errors; 13 training overlays; held-out labels untouched
Interventions	Nine views across evaluation-side, training-side, masking and combined changes

1. Dataset and Research Questions

The UCI SMS Spam Collection is a public corpus of 5,574 labelled SMS messages assembled from several sources for mobile-spam research [1]. The course manifest maps every 1-based UCI row to a stable ID and a fixed train or held-out split [2]. The raw archive and extracted corpus were SHA-256 checked, then joined one-to-one with the manifest. All row-count, ID, split-prefix, label-domain and empty-text assertions passed.

Split	Ham	Spam	Total	Spam share
Train	3,862	598	4,460	13.41%
Held-out	965	149	1,114	13.38%
All	4,827	747	5,574	13.40%

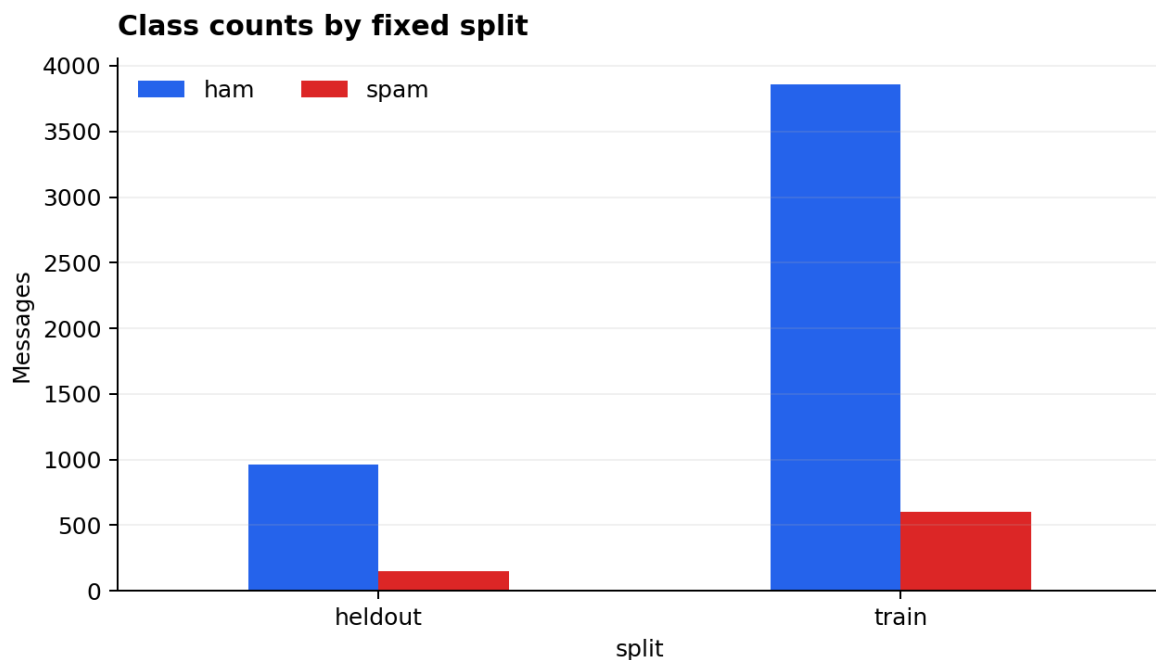


Figure 1. Class counts are closely matched in proportion across the fixed split.

Research questions

- (1) Are held-out examples independent of training?
- (2) Which labels have enough independent evidence to merit review?
- (3) How much performance can shallow features explain?
- (4) Which examples remain policy-ambiguous?
- (5) How do controlled data interventions change the meaning of evaluation?

2. Methodology

2.1 Profiling and baselines

The canonical table contains stable ID, split, 1-based UCI row number, raw text and public label. Derived features include message length, token count, digits, capitalization, URL, phone, currency, promotional tokens and normalized row position. The primary model is word TF-IDF over unigrams and bigrams with Logistic Regression. A character 3-5-gram model supplies a representation-diverse diagnostic. Training evidence uses five-fold out-of-fold probabilities; held-out models are fitted only on training [3].

2.2 Duplicates and leakage

Exact duplicates use the public lowercase-plus-whitespace-collapse rule. Near candidates are retrieved with word and character TF-IDF cosine search. A pair is eligible at a primary cosine of at least 0.92 and must not be exact. Cross-split candidates then receive a separate text-comparison decision. The audit retains rejected threshold crossings so precision control is visible.

2.3 Label errors and ambiguity

Label-error retrieval combines OOF or held-out model disagreement, representation-diverse model agreement, nearest-neighbour label evidence and cluster evidence. Final claims require at least two evidence sources, including a policy reading of the original text. Ambiguity is used only when missing consent or sender context leaves two plausible labels; low model confidence alone is rejected as proof.

2.4 Shortcut and intervention tests

Shallow Logistic Regression conditions isolate shape, contact/promotion and row-position features. A masked-text model replaces a predeclared promotional lexicon. Controlled interventions keep the word TF-IDF plus Logistic Regression pipeline fixed while changing leakage partners, a 13-row training-label overlay, ambiguous rows, shortcut cues, duplicate weighting, or the leakage-aware evaluation subset. No held-out label is edited.

Confidence calibration

High confidence requires direct definitional evidence and no unresolved counterexample. Medium confidence marks threshold- or policy-sensitive cases. Low confidence records rejected or weak candidates. The final ranked table contains 36 claims - six per issue type - with 19 high-confidence rows (52.8%), below the public 55% cap.

3. Dataset Profile and Baseline

Spam is longer and much more numeric than ham: median length is 149 versus 52 characters, median digit count is 16 versus 0, phone-like numbers occur in 60.78% versus 0.10%, and currency cues in 38.15% versus 0.62%. These gaps make strong shallow performance plausible before any lexical model is trained.

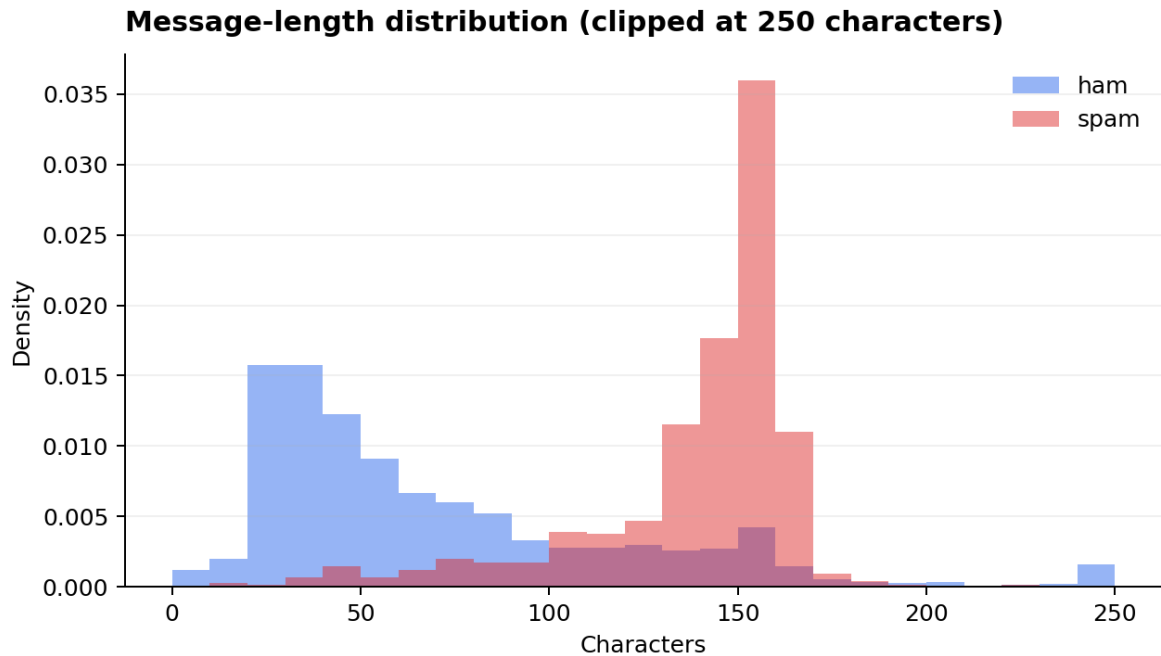


Figure 2. Spam messages are substantially longer than ham messages.

Condition	Accuracy	Spam P	Spam R	Spam F1	Macro F1
Word, train OOF	98.21%	99.24%	87.29%	92.88%	95.93%
Character, train OOF	98.27%	99.43%	87.63%	93.16%	96.08%
Word, held-out	99.10%	99.29%	93.96%	96.55%	98.02%
Character, held-out	98.74%	99.27%	91.28%	95.10%	97.19%

Raw held-out accuracy is high, but nine of ten errors are public spam rows predicted ham. Accuracy is therefore less informative than spam recall and audit-aware error analysis.

4. Duplicate and Leakage Findings

Lowercase and whitespace normalization yields 5,159 unique texts. The audit finds 290 exact clusters containing 705 rows; the largest is a 30-row training cluster. Five exact clusters cross the fixed split, all with consistent labels. Their problem is independence, not label conflict.

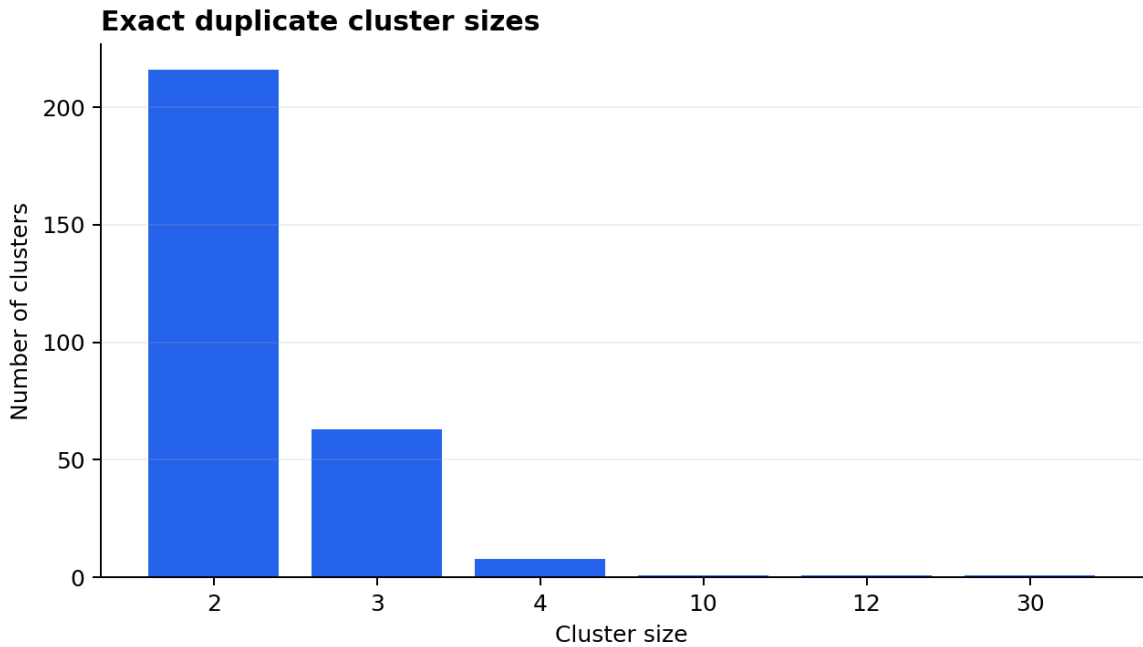


Figure 3. Most exact clusters are small, but a few repeated templates are large.

Threshold	All pairs	Cross-split	Label conflicts
0.88	447	129	0
0.90	405	108	0
0.92	365	90	0
0.94	313	56	0
0.96	275	46	0

At 0.92, 365 non-exact pairs remain and 90 cross the split. After excluding 17 heldout-heldout pairs, the 348-pair review queue accepts 200 pairs and rejects 148 false positives. Eighty-five accepted pairs cross the split and affect 60 unique held-out rows. Rejections are mostly generic conversational phrases that share wording without a specific message or campaign anchor.

5. Leakage-Aware Evaluation

The accepted relationships affect five exact held-out rows and 60 near-matched held-out IDs. Their union contains 63 rows because two held-out messages participate in both exact and near relationships. Leakage-aware evaluation keeps the fitted model fixed and removes those rows only from scoring; the training-side intervention separately removes 71 training partners and retrains.

Condition	Train N	Test N	Accuracy	Spam recall	Spam F1
Original public split	4,460	1,114	99.10%	93.96%	96.55%
Leak-aware evaluation	4,460	1,051	99.05%	90.91%	94.74%
Remove train partners	4,389	1,114	99.01%	92.62%	96.17%

Overall accuracy falls by only 0.05 points, while spam recall falls by 3.05 points and spam F1 by 1.81 points. The minority-class metrics reveal a meaningful novelty penalty that the headline accuracy conceals.

The result does not prove that every repeated template is invalid in deployment. Recurring spam campaigns can be realistic. It does prove that the fixed held-out set mixes novel examples with messages recoverable from training, so a single score cannot support both file-level and novel-template generalization claims.

6. Likely Label Errors and Ambiguity

Automated retrieval produced 70 label-error candidates. Evidence-and-policy adjudication retains 15: five high-confidence, nine medium-confidence and one low-confidence. Thirteen training proposals are stored in an overlay; two held-out cases remain flags only. All fifteen are public spam rows proposed as ham, and the clearest cases are standalone jokes with no solicitation, charge or response mechanism.

ID	Public	Proposed	Confidence	Reason
T2378	spam	ham	High	Driving joke; both OOF models strongly ham
H0143	spam	ham-like	High	Tattoo joke; held-out label unchanged
T0059	spam	ham	High	Divorce Barbie joke; no action or charge
T2232	spam	ham	High	Child/teenager joke
H0896	spam	ham-like	High	Victim query; held-out label unchanged
T3272	spam	ham	Medium	Personal debt-collector reply

Fourteen ambiguity claims are reported separately: 13 medium and one low confidence. They cover adult-chat style, 070-number follow-up, research recruitment, charity, club advertising, service onboarding and truncated context. H0935 and T0470 are retained as explicit rejected candidates: their models are uncertain, but the messages are ordinary ham. Uncertainty alone is not ambiguity.

Two of the baseline's ten held-out errors are likely label-policy errors. Treating them as unresolved changes an audit sensitivity accuracy from 99.10% to 99.28%; this is not presented as a corrected official score.

The policy review was AI-assisted and is not represented as two independent human annotators. No fabricated kappa or agreement statistic is reported. This disclosure limits the strength of subjective sample-level conclusions while preserving reproducibility of the automated evidence.

7. Shortcut Features

Contact and promotion features have substantial predictive power. Digit count has the largest standardized shallow-model coefficient (2.434), followed by promotional-token count (1.671), average token length (0.740), URL presence (0.666) and currency count (0.636). Row position adds no measurable value.

Condition	Accuracy	Spam recall	Spam F1	Macro F1
Majority ham	86.62%	0.00%	0.00%	46.42%
Shape only	89.50%	37.58%	48.91%	71.53%
Contact/promotion	98.11%	90.60%	92.78%	95.85%
All shallow	98.20%	91.28%	93.15%	96.06%
Masked text	98.92%	93.96%	95.89%	97.64%
Full text	99.10%	93.96%	96.55%	98.02%

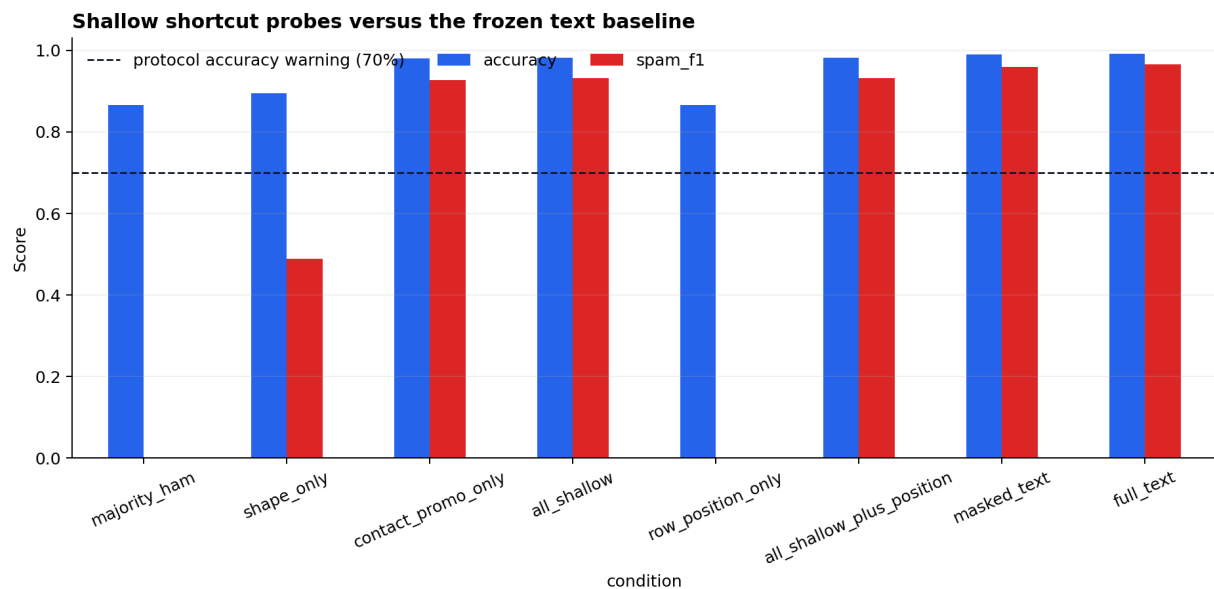


Figure 4. Shallow features are strong, but their example-level failures are systematic.

The cues are relevant but fragile. H0946 is a legitimate price discussion assigned shallow spam probability 0.992; H0044 is a ringtone solicitation assigned shallow probability 0.085. Premium-rate adult-chat messages can be caught almost entirely by price and phone markers even when the full text classifier misses them.

8. Data-Intervention Results

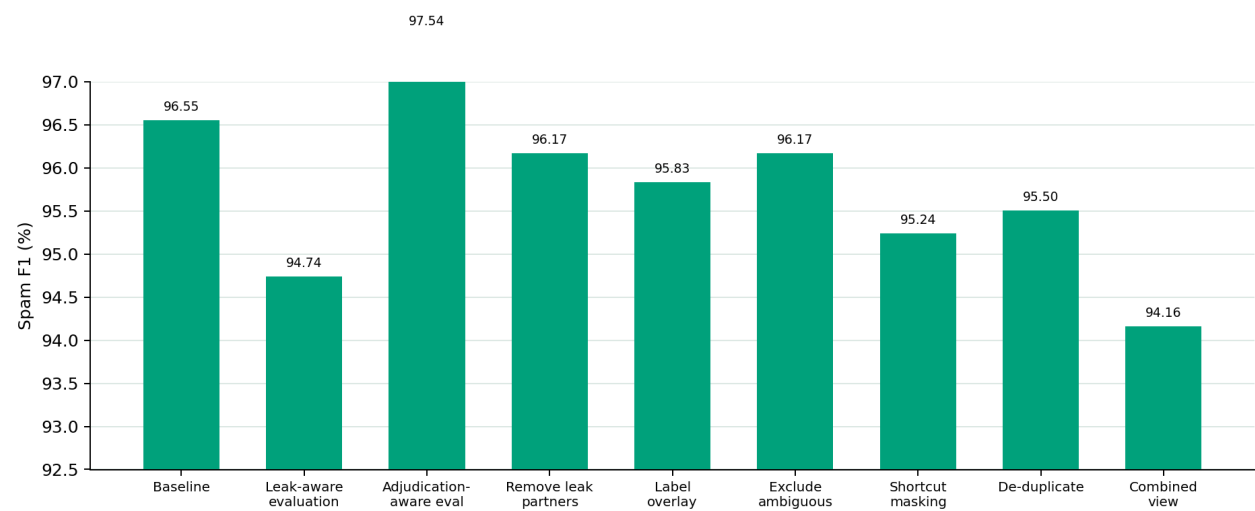


Figure 5. Data interventions change minority-class metrics more than headline accuracy.

Condition	Train N	Test N	Accuracy	Spam recall	Spam F1
Original	4,460	1,114	99.10%	93.96%	96.55%
Leak-aware eval	4,460	1,051	99.05%	90.91%	94.74%
Adjudication-aware	4,460	1,110	99.37%	95.86%	97.54%
Remove leak partners	4,389	1,114	99.01%	92.62%	96.17%
Label overlay	4,460	1,114	98.92%	92.62%	95.83%
Exclude ambiguous	4,448	1,114	99.01%	92.62%	96.17%
Shortcut masking	4,460	1,114	98.74%	93.96%	95.24%
De-duplicate	4,043	1,114	98.83%	92.62%	95.50%
Combined view	3,982	1,114	98.47%	91.95%	94.16%

Duplicate de-duplication removes 417 repeated training rows; the combined view removes 478 rows before the label overlay and masking steps. The 13-row label overlay changes only two held-out predictions. Adjudication-aware evaluation moves in the opposite direction, but three of its four excluded rows were baseline errors and the suspected-label subset was retrieved using model opposition. The experiments therefore discourage score-driven cleaning: direction alone is not evidence that an intervention is correct.

9. Difficulties and Solutions

Difficulty 1 - Stable raw-row alignment

A single parse offset would invalidate every stable ID and hidden-key comparison. The solution was to split each raw line only on its first tab, use a 1-based row number, perform a one-to-one merge, and stop on any count, ID, split-prefix, label-domain or empty-text failure. Verification is stored in `data_validation.json`.

Difficulty 2 - TF-IDF false-positive traps

High cosine similarity can still join generic phrases or texts with a meaningful semantic change. We kept the public threshold for reproducibility, added character similarity and a separate text-comparison decision, and retained five rejected cross-split pairs. The accepted-versus-rejected ledger makes the precision control directly auditable.

Difficulty 3 - Label error versus model error

The conservative classifier produces many spam false negatives, most of which remain valid spam. We required two evidence sources and a policy reading, then stored training proposals as an overlay rather than editing public labels. Seventy automated candidates were reduced to fifteen submitted findings.

Difficulty 4 - Accuracy conceals minority-class impact

Leakage-aware accuracy changes by only 0.05 points, which initially appears negligible. Reporting spam recall and F1 reveals a 3.05-point recall decline. The solution was to predefine class-specific metrics and interpret score decreases as potentially stricter evaluation rather than model failure.

Difficulty 5 - Honest AI-assisted adjudication

The project requires manual judgment, but this implementation was independently executed with AI assistance. We separated automated evidence from a policy pass, logged every selected judgment, and explicitly avoided claiming two human reviewers or fabricated agreement statistics. Independent team sign-off remains advisable before academic submission.

10. Conclusions and Recommendations

The fixed SMS corpus supports a strong classifier, but the 99.10% headline accuracy is not a complete generalization statement. Five exact clusters cross the split, 60 held-out IDs participate in accepted near matches, shallow cues nearly match the text model, and a small number of policy-sensitive labels account for a material share of observed errors.

Recommended benchmark redesign: group exact and accepted near templates before splitting; publish a consent-aware annotation guide; preserve an uncertain/policy-boundary category; and report spam recall, spam F1, and leakage-aware sensitivity beside overall accuracy.

The most defensible quantitative conclusion is narrow: overall accuracy is stable, but novel-template spam recall is lower than the raw split suggests. The audit does not claim definitive ground truth for every subjective message, and it does not treat performance changes as proof that a cleaning decision is correct.

AI usage declaration

OpenAI Codex was used to interpret the supplied specification, implement and test the analysis pipeline, generate candidate evidence, perform an explicitly AI-assisted policy pass, draft documentation, and generate this report. AI outputs were treated as untrusted drafts: row mappings were checked by assertions, numerical claims were regenerated from saved CSV files, threshold traps were compared against original text, and submission constraints were checked programmatically. No claim of independent human double-review is made, and AI is not presented as a source of ground-truth labels. The complete usage record is in logs/chat.md.

References

- [1] T. Almeida and J. Hidalgo, SMS Spam Collection, UCI Machine Learning Repository, 2011. DOI: 10.24432/C5CC84. License: CC BY 4.0.
- [2] AIAA3102 Final Project Topic B: Dataset Forensics and Label Audit, course handout and starter protocol supplied with this repository, 2026.
- [3] F. Pedregosa et al., Scikit-learn: Machine Learning in Python, Journal of Machine Learning Research 12, 2825-2830, 2011.

Reproducibility: from the repository root, run the data foundation scripts first, then `audit_duplicates.py`, `review_near_duplicates.py`, `audit_leakage.py`, `audit_label_noise.py`, `audit_shortcuts.py`, `run_impact_interventions.py`, `build_final_submission.py`, and finally `generate_report.py`. The public raw archive is downloaded on demand and checksum-recorded; seed 42 is fixed in model experiments.